Generating Functions

Let $a_0, a_1, a_2, \ldots$ be a sequence of real numbers. The function

$$f(x) = a_0 + a_1 x + a_2 x^2 + \cdots$$
$$= \sum_{i=0}^{\infty} a_i x^i$$

is called the generating function for the given sequence.

① Obtain the generating function for the sequence $\{1, 1, 1, \ldots\}$.

▲
$$f(x) = 1 + x + x^2 + \cdots = (1-x)^{-1} = \frac{1}{1-x}$$

$$\therefore \boxed{f(x) = \frac{1}{1-x}}$$

② -11- $1, -1, 1, -1, 1, -1, \ldots$

▲
$$f(x) = 1 - x + x^2 - x^3 + x^4 - x^5 + \cdots = (1+x)^{-1} = \frac{1}{1+x}$$

$$\therefore \boxed{f(x) = \frac{1}{1+x}}$$

③ Obtain the generating function for the sequence of Natural numbers.

▲ We have $N = \{1, 2, 3, \ldots\}$

$$f(x) = 1 + 2x + 3x^2 + 4x^3 + \cdots$$
$$= D[x + x^2 + x^3 + x^4 + \cdots]$$
$$= D\,x[1 + x + x^2 + x^3 + \cdots]$$
$$= D\left[\frac{x}{1-x}\right]$$
$$= \frac{(1-x)(1) - x(-1)}{(1-x)^2} = \frac{1}{(1-x)^2}$$

$$\therefore \boxed{f(x) = \frac{1}{(1-x)^2}}$$

(4). Obtain the generating function for the sequence $\{0, 1, 1, 1, \ldots\}$

▲ $f(x) = 0 + x + x^2 + x^3 + \cdots$

$= x[1 + x + x^2 + \cdots] = \frac{x}{1-x}$

$$\boxed{f(x) = \frac{x}{1-x}}$$

(5) Obtain the generating function for the sequence $\{0, 1, -1, 1, -1, 1, \ldots\}$.

▲ $f(x) = 0 + x - x^2 + x^3 - x^4 + x^5 - \cdots$

$= x[1 - x + x^2 - x^3 + x^4 - \cdots] = \frac{x}{1+x}$

$$\therefore \boxed{f(x) = \frac{x}{1+x}}$$

(6). Obtain the generating function for the sequence of cubes of natural numbers.

▲ Given that $1^3, 2^3, 3^3, \ldots\ldots$

$f(x) = 1^3 + 2^3 x + 3^3 x^2 + \cdots$

$= 1^2 D(x) + 2^2 D[x^2] + 3^2 D[x^3] + \cdots$

$= D[x + 2^2 x^2 + 3^2 x^3 + \cdots]$

$= D(x)[1 + 2^2 x + 3^2 x^2 + \cdots]$

$= D(x)[D[x + 2x^2 + 3x^3 + \cdots]]$

$= D(x)[D(x)[1 + 2x + 3x^2 + \cdots]]$

$= D(x)\left[D(x)\frac{1}{(1-x)^2}\right]$

$= D(x)\left[\frac{[1-x]^2(1) - x\,2[1-x](-1)}{[1-x]^4}\right]$

$= D(x)\left[\frac{[1-x] + 2x}{[1-x]^3}\right]$

$= D(x)\left[\frac{1+x}{[1-x]^3}\right]$

$= D\left[\frac{x(1+x)}{[1-x]^3}\right]$

$$= \frac{[1-x]^3[1+2x] - x[1+x] \times 3[1-x]^2(-1)}{[1-x]^6}$$

$$= \frac{[1-x][1+2x] + 3x[1+x]}{[1-x]^4}$$

$$= \frac{1+2x-x-2x^2+3x+3x^2}{[1-x]^4}$$

$$= \frac{1+4x+x^2}{[1-x]^4}$$

$$\therefore f(x) = \frac{1+4x+x^2}{[1-x]^4}$$

(7). Find the generating function for

$$a_r = \begin{cases} 3^r, & \text{if } r \text{ is Even} \\ -3^r, & \text{if } r \text{ is odd} \end{cases}$$

▲ $a_0 = 1,\ a_1 = -3,\ a_2 = 3^2,\ a_3 = -3^3, \ldots$

$$f(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + \cdots$$

$$f(x) = 1 - 3x + 3^2x^2 - 3^3x^3 + \cdots$$

$$f(x) = 1 - (3x) + (3x)^2 - (3x)^3 + \cdots$$

$$f(x) = 1 - X + X^2 - X^3 + \cdots, \text{ where } X = 3x$$

$$f(x) = \frac{1}{1+X} \Rightarrow \boxed{f(x) = \frac{1}{1+3x}}$$

(8). Find the generating function for the sequence

0, 2, 6, 12, 20, 30, 42, ……

▲ $f(x) = 0 + 2x + 6x^2 + 12x^3 + \cdots$

$$f(x) = [0+0] + [1+1]x + [2+4]x^2 + [3+9]x^3 + \cdots$$

$$f(x) = [0 + x + 2x^2 + 3x^3 + \cdots] + [0 + x + 4x^2 + 9x^3 + \cdots]$$

$$= x[1 + 2x + 3x^2 + \cdots] + x[1 + 2^2x + 3^2x^2 + \cdots]$$

$$= \frac{x}{[1-x]^2} + x\,D[x + 2x^2 + 3x^3 + \cdots]$$

$$f(x) = \frac{x}{[1-x]^2} + x\,D(x)\,[1 + 2x + 3x^2 + \cdots]$$

$$f(x) = \frac{x}{[1-x]^2} + x\,D\left[\frac{x}{[1-x]^2}\right]$$

$$f(x) = \frac{x}{[1-x]^2} + x\left[\frac{[1-x]^2(1) - x\;2[1-x](-1)}{[1-x]^4}\right]$$

$$f(x) = \frac{x}{[1-x]^2} + x\left[\frac{[1-x] + 2x}{[1-x]^3}\right]$$

$$f(x) = \frac{x}{[1-x]^2} + \frac{x[1+x]}{[1-x]^3}$$

$$f(x) = \frac{x[1-x] + x[1+x]}{[1-x]^3} = \frac{x - x^2 + x + x^2}{[1-x]^3}$$

$$\therefore f(x) = \frac{2x}{[1-x]^3}$$

(9). Find the generating function for the sequence

{ 1, 1, 1, 1, 1, 0, 1, 1, 0, 1, 1, ... }

▲ From the given given sequence, we have

$a_5 = 0$ and $a_8 = 0$.

$$f(x) = 1 + x + x^2 + x^3 + x^4 + (0x^5) + x^6 + x^7 + [0x^8] + x^9 + x^{10} + \cdots\cdots$$

$$\therefore f(x) = \frac{1}{1-x} - x^5 - x^8$$

(10). Find the generating function for the sequence

0, 1, 3, 9, 27, ...

▲ $f(x) = 0 + x + 3x^2 + 9x^3 + 27x^4 + \cdots$

$$f(x) = x + 3^2x^2 + 3^3x^3 + 3^4x^4 + \cdots$$

$$= x\,[1 + 3^2x + 3^3x^2 + 3^4x^3 + \cdots]$$

$$= x\,D\,[x + 3x^2 + 3^2x^3 + 3^3x^4 + \cdots]$$

$$= x\,D\,x\,[1 + 3x + 3^2x^2 + 3^3x^3 + \cdots]$$

$$= x\,D\,x\,[1 + X + X^2 + X^3 + \cdots], \text{ where } X = 3x$$

$$= x\,D\,x \times \frac{1}{1-X} = x\,D(x)\left[\frac{1}{1-3x}\right]$$

$$f(x) = x\, D\, x\left[\frac{1}{1-3x}\right]$$

$$f(x) = x\, D\left[\frac{x}{1-3x}\right] = x\left[\frac{[1-3x](1) - x(-3)}{[1-3x]^2}\right]$$

$$f(x) = x\left[\frac{1-3x+3x}{[1-3x]^2}\right]$$

$$\therefore\ f(x) = \frac{x}{[1-3x]^2}$$

1. Find the generating function for the sequence

$\binom{8}{0}, \binom{8}{1}, \binom{8}{2}, \ldots, \binom{8}{8}$.

Given that $a_0 = \binom{8}{0}$, $a_1 = \binom{8}{1}$, $a_2 = \binom{8}{2}$, $\ldots$ $a_8 = \binom{8}{8}$, $a_9 = 0$, $a_{10} = 0 \ldots$

$$f(x) = {}^8C_0 + \binom{8}{1}x + \binom{8}{2}x^2 + \cdots + \binom{8}{8}x^8$$

$$f(x) = [1+x]^8$$

2. Find the generating function, for the sequence

$\binom{8}{1}, 2\binom{8}{2}, 3\binom{8}{3}, \ldots, 8\binom{8}{8}$.

Given that $a_0 = \binom{8}{1}$, $a_1 = 2\binom{8}{2}$, $a_2 = 3\binom{8}{3}$, $\ldots$ $a_7 = 8\binom{8}{8}$

$$f(x) = \binom{8}{1} + 2\binom{8}{2}x + 3\binom{8}{3}x^2 + \cdots + 8\binom{8}{8}x^7$$

$$f(x) = D\left[\binom{8}{1}x + \binom{8}{2}x^2 + \binom{8}{3}x^3 + \cdots + \binom{8}{8}x^8\right]$$

$$f(x) = D\left[{}^8C_0 + \binom{8}{1}x + \binom{8}{2}x^2 + \binom{8}{3}x^3 + \cdots \binom{8}{8}x^8 - 1\right]$$

$$= D\left[{}^8C_0 + \binom{8}{1}x + \binom{8}{2}x^2 + \binom{8}{3}x^3 + \cdots + \binom{8}{8}x^8\right] - D[1]$$

$$= D\,[1+x]^8 - 0$$

$$= 8\,[1+x]^7$$

$$\therefore\ f(x) = 8\,[1+x]^7$$

(13) Find the generating function, for the sequence

$\binom{n}{0}, \binom{n}{1}, \binom{n}{2}, \cdots, \binom{n}{n}, 0, 0, 0, 0, \cdots$

▲ $f(x) = \binom{n}{0} + \binom{n}{1}x + \binom{n}{2}x^2 + \cdots + \binom{n}{n}x^n + 0\,x^{n+1} + \cdots$

$f(x) = [1+x]^n$.

(14). Repeated Find the generating function, for the sequence

1, 2, 3, 4, ----

▲ $f(x) = 1 + 2x + 3x^2 + 4x^3 + \cdots$

$= D[x + x^2 + x^3 + x^4 + \cdots]$

$= D\,x[1 + x + x^2 + x^3 + \cdots]$

$= D\left[\frac{x}{1-x}\right] = \frac{[1-x](1) - x(-1)}{[1-x]^2} = \frac{1}{[1-x]^2}$

$\therefore f(x) = \frac{1}{[1-x]^2}$

(15) Find the generating function, for the sequence of ~~Square of natural numbers~~ $0^2, 1^2, 2^2, 3^2, \cdots$

▲ $f(x) = 1^2x + 2^2x^2 + 3^2x^3 + \cdots$

$= x + [illegible]$

$= [illegible]$

$= x[1 + 2^2x + 3^2x^2 + \cdots]$

$= x\,D[x + 2x^2 + 3x^3 + \cdots]$

$= x\,D(x)[1 + 2x + 3x^2 + \cdots]$

$= x\,D\left[\frac{x}{[1-x]^2}\right]$

$= x\left[\frac{[1-x]^2(1) - x\,2(1-x)(-1)}{[1-x]^4}\right]$

$= x\left[\frac{[1-x] + 2x}{[1-x]^3}\right]$

$= \frac{x(1+x)}{[1-x]^3}$

$\therefore f(x) = \frac{x(1+x)}{[1-x]^3}$

Find the generating function for the sequence of

1, 0, 1, 0, 1, 0, 1, ---

$f(x) = 1 + x^2 + x^4 + x^6 + \cdots = \frac{1}{1-x^2}$

$\therefore f(x) = \frac{1}{1-x^2}$

(17) Find the generating function, for the sequence of

$0, 0, 0, 6, -6, 6, -6, 6, \ldots$

▶ $f(x) = [0 + 0 \times x + 0 \times x^2 + 6x^3 - 6x^4 + 6x^5 - 6x^6 + 6x^7 - \ldots]$

$= 6x^3 [1 - x + x^2 - x^3 + x^4 - \ldots]$

$= \frac{6x^3}{1-x}$

$\therefore f(x) = \frac{6x^3}{1-x}$

(18) Find the generating function, for the sequence of

$0, 0, 1, a, a^2, a^3, \ldots, \quad a \neq 0$.

▶ $f(x) = 0 + 0x + x^2 + ax^3 + a^2x^4 + a^3x^5 + \ldots$

$= x^2 [1 + ax + a^2x^2 + a^3x^3 + \ldots]$

$= x^2 \times \frac{1}{1-ax}$

$\therefore f(x) = \frac{x^2}{1-ax}$

(19) In each of the following, $f(x)$ is a generating function for the sequence $\langle a_r \rangle$ and $g(x)$ is a generating function for the sequence $\langle b_r \rangle$. Express $g(x)$ in terms of $f(x)$.

(i) $b_3 = 3$; $b_7 = 7$. $b_n = a_n$, for $n \neq 3, 7$.

(ii) $b_1 = 1$, $b_3 = 3$, $b_7 = 7$, $b_n = 2a_n + 5$, for $n \neq 1, 3, 7$.

▶ We have

(i) $g(x) = b_0 + b_1 x + b_2 x^2 + b_3 x^3 + b_4 x^4 + b_5 x^5 + b_6 x^6 + b_7 x^7 + b_8 x^8 + \cdots$

$= a_0 + a_1 x + a_2 x^2 + 3x^3 + a_4 x^4 + a_5 x^5 + a_6 x^6 + 7x^7 + a_8 x^8 + \cdots$

$g(x) = f(x) - a_3 x^3 - a_7 x^7 + 3x^3 + 7x^7$

$= f(x) + [3 - a_3] x^3 + [7 - a_7] x^7$.

(ii) $g(x) = \sum_{r=0}^{\infty} b_r x^r$

$= \left\{ \sum_{r=0}^{\infty} [2a_r + 5] x^r - [2a_1 + 5] x - [2a_3 + 5] x^3 - [2a_7 + 5] x^7 \right\} + b_1 x + b_3 x^3 + b_7 x^7$.

$= 2 \sum_{r=0}^{\infty} a_r x^r + 5 \sum_{r=0}^{\infty} x^r + [b_1 - 2a_1 - 5] x + [b_3 - 2a_3 - 5] x^3 + [b_7 - 2a_7 - 5] x^7$

$g(x) = 2 f(x) + 5[1-x]^{-1} - 2[a_1 + 2]x - 2[a_3 + 1] x^3 - 2(a_7 - 1) x^7$.

(20) Find the generating function for the following sequence

(i) $1^2, 2^2, 3^2, \cdots$ (ii) $0^2, 1^2, 2^2, \cdots$ (iii) $1^3, 2^3, 3^3, 4^3, \cdots$

(iv) $0^3, 1^3, 2^3, \cdots$

(i) $f(x) = \frac{1+x}{(1-x)^3}$ (ii) $f(x) = \frac{x(1+x)}{(1-x)^3}$ (iii) $f(x) = \frac{x^3 + 4x + 1}{[1-x]^4}$

(iv) $f(x) = \frac{x(x^2 + 4x + 1)}{(1-x)^4}$

(21) Find a the generating function for each of the following sequence

(i) $1, 1, 0, 1, 1, 1, \cdots$ (ii) $0, 2, 6, 12, 20, 30, 42, \cdots$

(iii) $8, 26, 54, 92, \cdots$

(i) $f(x) = \frac{1}{1-x} - x^2$ (iii) $f(x) = (0+0) + (1+1)x + [2+4]x^2 + [3+9]x^3 + \cdots$

$f(x) = [x + 2x^2 + 3x^3 + \cdots] + [x + 4x^2 + 9x^3 + \cdots]$

$f(x) = x[1 + 2x + 3x^2 + \cdots] + x[1 + 4x + 9x^2 + \cdots]$

$f(x) = \frac{x}{(1-x)^2} + x D[x + 2x^2 + 3x^3 + \cdots]$

$f(x) = \frac{x}{(1-x)^2} + x\,D(x)\,[1 + 2x + 3x^2 + \cdots]$

$f(x) = \frac{x}{(1-x)^2} + x\,D(x)\left[\frac{1}{(1-x)^2}\right] = \frac{x}{(1-x)^2} + \frac{x(1+x)}{(1-x)^3} = \frac{2x}{(1-x)^3}$.

(iii) $f(x) = 8 + 26x + 54x^2 + 92x^3 + \cdots$

$8, 26, 54, 92, \cdots$

$a_r = 3(r+1) + 5(r+1)^2, \quad r = 0, 1, 2, \cdots$

$\langle r+1 \rangle = 1, 2, 3, \cdots$

$\langle (r+1)^2 \rangle = 1^2, 2^2, 3^2, \cdots$

$f(x) = \frac{3}{(1-x)^2} + \frac{5(1+x)}{(1-x)^3} = \frac{8+2x}{(1-x)^3}$.

(22) Find the generating function for the sequence $\langle a_r \rangle$, where $a_r = 1$, for $0 \le r \le n$ and $a_r = 0$, for $r \ge n+1$

▲ We have $\frac{1}{1-x} = 1 + x + x^2 + x^3 + \cdots + x^n + x^{n+1} + x^{n+2} + \cdots$

Note :- ① $1 + x + x^2 + \cdots + x^n = \frac{1-x^{n+1}}{(1-x)}$

② For common difference $a_n = a_1 + (n-1)d$

(23) Prove that $1 + x + x^2 + \cdots + x^n = \frac{1-x^{n+1}}{1-x}$

▲ $1 + x + x^2 + \cdots + x^n + x^{n+1} + \cdots - [x^{n+1} + x^{n+2} + \cdots]$

$= \frac{1}{1-x} - x^{n+1}[1 + x + x^2 + \cdots]$

$= \frac{1}{1-x} - \frac{x^{n+1}}{1-x} = \frac{1-x^{n+1}}{1-x}$.

① Find the sequence generated by the function $[1+x]^n$.

▶ Given that $f(x) = [1+x]^n$

$= {}^nC_0 + {}^nC_1 x + {}^nC_2 x^2 + \cdots + {}^nC_n x^n$.

The sequence of $[1+x]^n$ is ${}^nC_0, {}^nC_1, \cdots, {}^nC_n, 0, 0, 0, \cdots$

② Find the sequence generated by the function $\frac{1-x^{n+1}}{1-x}$.

▶ Given that $f(x) = \frac{1-x^{n+1}}{1-x}$

$= 1 + x + x^2 + \cdots + x^n$

The sequence of $\frac{1-x^{n+1}}{1-x}$ is

$1, 1, 1, \cdots, 1, 0, 0, 0, \cdots$, where first $n+1$ terms are '1'.

③, ④ $\frac{x}{[1-x]^2}$.

▶ Given that $f(x) = \frac{x}{(1-x)^2}$

$= x[1 + 2x + 3x^2 + 4x^3 + \cdots]$

$= x + 2x^2 + 3x^3 + 4x^4 + \cdots$

$= 0 + x + 2x^2 + 3x^3 + 4x^4 + \cdots$

The sequence of $\frac{x}{(1-x)^2}$ is

$0, 1, 2, 3, 4, \cdots$.

⑤ $\frac{1}{[1-x]^2}$.

▶ $1, 2, 3, 4, \cdots$

⑤ Find the sequence generated by the function

$f(x) = \frac{x+1}{(1-x)^3}$

▶ We have $\frac{x}{[1-x]^2} = x[1 + 2x + 3x^2 + 4x^3 + \cdots]$

$= [x + 2x^2 + 3x^3 + 4x^4 + \cdots]$

$\frac{d}{dx}\left[\frac{x}{[1-x]^2}\right] = \frac{d}{dx}[x + 2x^2 + 3x^3 + 4x^4 + \cdots]$

$\frac{x+1}{[1-x]^3} = [1 + 2^2 x + 3^2 x^2 + 4^2 x^3 + \cdots]$

∴ The sequence of given g.f is $1^2, 2^2, 3^2, \cdots$

(6). Find the sequence generated by the function

$$\frac{x(x+1)}{(1-x)^3}$$

▲ We have $\frac{x+1}{(1-x)^3} = 1^2 + 2^2 x + 3^2 x^2 + 4^2 x^3 + \cdots$

$$\frac{x(x+1)}{(1-x)^3} = 1^2 x + 2^2 x^2 + 3^2 x^3 + 4^2 x^4 + \cdots$$

The sequence given g.f is

$0, 1^2, 2^2, \cdots$

(7). ——— $\frac{1}{1-2x}$

$$\frac{1}{1-2x} = 1 + (2x) + (2x)^2 + (2x)^3 + \cdots$$

The sequence of given g.f is

$1, 2, 2^2, 2^3, \cdots$ (or) $2^0, 2^1, 2^2, 2^3, \cdots$

(8). ——— $\frac{1}{1-x} + 3x$

▲ $f(x) = \frac{1}{1-x} + 3x = 1 + 4x + x^2 + x^3 + x^4 + \cdots$

$\therefore$ $1, 4, 1, 1, \cdots$

(9). ——— $e^x + 2x^3$

▲ $f(x) = e^x + 2x^3 = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots + 2x^3$

$$= 1 + \frac{x}{1!} + \frac{x^2}{2!} + \left[\frac{1}{3!} + 2\right] x^3 + \cdots$$

$\therefore$ $1, 1, \frac{1}{2!}, \left[2 + \frac{1}{3!}\right], \frac{1}{4!}, \cdots$

Generating function

3. Determine the sequence generated by each of the following generating functions:

(i) $f(x) = [2x-3]^3$ (ii) $f(x) = \frac{x^4}{1-x}$ (iii) $f(x) = \frac{x^3}{1-x^2}$

(iv) $f(x) = \frac{1}{1+3x}$ (v) $f(x) = \frac{1}{3-x}$ (vi) $\frac{1}{1-x} + 3x^7 - 11$

▶ (i) $-27, 54, -36, 8, 0, 0, 0 \cdots$

(ii) $\frac{x^4}{1-x} = x^4[1 + x + x^2 + x^3 + \cdots] = x^4 + x^5 + x^6 + \cdots$

$\therefore 0, 0, 0, 0, 1, 1, \cdots$

(iii) $\frac{x^3}{1-x^2} = x^3[1 + (x^2) + [x^2]^2 + (x^2)^3 + \cdots]$

$= x^3 + x^5 + x^7 + x^8 + x^9 + \cdots$

$\therefore 0, 0, 0, 1, 0, 1, 0, 1, 1, 1, \cdots$

(iv) $\frac{1}{1+3x} = 1 - (3x) + (3x)^2 - (3x)^3 + \cdots$

$\therefore 1, -3, 3^2, -3^3, \cdots$

(v) $\frac{1}{3-x} = \frac{1}{3\left[1 - \frac{x}{3}\right]}$

$= \frac{1}{3}\left[1 + \left(\frac{1}{3}\right)x + \left(\frac{1}{3}\right)^2 x^2 + \cdots\right]$

$= \frac{1}{3} + \left(\frac{1}{3}\right)^2 x + \left(\frac{1}{3}\right)^3 x^2 + \cdots$

$\therefore \frac{1}{3}, \left(\frac{1}{3}\right)^2, \left(\frac{1}{3}\right)^3, \cdots$

(vi) $\frac{1}{1-x} + 3x^7 - 11 = -10 + x + x^2 + x^3 + x^4 + x^5 + x^6 + 4x^7 + \cdots$

$\therefore -10, 1, 1, 1, 1, 1, 1, 4, 1, \cdots$

$a_0 = -10, \quad a_7 = 4, \quad a_i = 1, \quad \forall\, i \neq 0, 7.$

① Find the binomial coefficient, if (i) n is integer (i.e $n \in \mathbb{R}$) (ii) n is -ve integer.

(i) We have $n C_r = \frac{n!}{(n-r)!\, r!}$, where $n, r \in \mathbb{Z}^+$ and $n \geq r > 0$

$$n C_r = \frac{n(n-1)\cdots[n-(r-1)] \times (n-r)!}{(n-r)! \times r!} = \frac{n(n-1)(n-2)\cdots(n-r+1)}{r!}$$

$\therefore$ If $n \in \mathbb{R}$, then

$$n C_r = \frac{n(n-1)(n-2)\cdots(n-r+1)}{r!} \quad \text{— (1)}$$

(ii) From (1), we have

$$n C_r = \frac{n(n-1)(n-2)\cdots(n-r+1)}{r!}$$

Replace n by $-n$

$$-n C_r = \frac{-n(-n-1)(-n-2)\cdots(-n-r+1)}{r!}$$

$$= (-1)^r \frac{(n+r-1)\cdots(n+2)(n+1)n}{r!} \cdot \frac{(n-1)!}{(n-1)!}$$

$$-n C_r = (-1)^r \frac{[n+r-1]!}{(n-1)!\, r!} = (-1)^r (n+r-1) C_r$$

$\therefore$ $n \in \mathbb{Z}^-$, $-n C_r = (-1)^r \cdot C[n+r-1, r]$ $= (-1)^r \, {}^{n+r-1}C_r$

② Find the coefficient of x^5 in $[1-2x]^{-7}$.

$$[1-2x]^{-7} = [1+X]^{-7} = \sum_{r=0}^{\infty} (-1)^r \binom{n+r-1}{r} X^r, \text{ where } X = -2x$$

$$= \sum_{r=0}^{\infty} {}^{-7}C_r \, X^r$$

$$= \sum_{r=0}^{\infty} (-1)^r \binom{7+r-1}{r} X^r$$

Consequently, for $r = 5$, we have

$$(-1)^5 \binom{7+5-1}{5} x^5 = (-1)\binom{11}{5}(-2x)^5$$

$$= {}^{11}C_5 \, 2^5 x^5$$

$\therefore$ The coefficient of x^5 is ${}^{11}C_5 \times 2^5 = 14,784$.

③. Find the coefficient of x^6 in $[1-4x]^{-9}$.

▲ $[1-4x]^{-9} = [1+X]^{-9} = \sum_{r=0}^{\infty} -9c_r X^r$ where $X = -4x$

$= \sum_{r=0}^{\infty} (-1)^r \; 9+r-1 c_r \; X^r$

For $r=6$, we have

$(-1)^6 \; 9+6-1 c_6 \; x^6 = 14c_6 x^6 = 14c_6 (-4x)^6 = 14c_6 \times 4^6 \times x^6$

∴ The coefficient of x^6 is $14c_6 \times 4^6$.

④. ✓ Determine the coefficient of x^{15} in $f(x) = [x^2 + x^3 + x^4 + \cdots]^4$.

▲ $f(x) = [x^2 + x^3 + x^4 + \cdots]^4$

$= x^8 [1 + x + x^2 + \cdots]^4$

$= x^8 \frac{1}{[1-x]^4} = x^8 [1-x]^{-4}$

$= x^8 \sum_{r=0}^{\infty} -4c_r (-x)^r$

$= x^8 \sum_{r=0}^{\infty} (-1)^r \; 4+r-1 c_r \; x^r$

For $r=7$, we have $x^8 (-1)^7 \; 4+7-1 c_7 (-x)^7$

$= 10c_7 \; x^{15}$

∴ The coefficient of x^{15} is $(+1) \times 10c_7 = +120$.

⑤. ——— 11 ——— x^{14} in $f(x) = [x^2 + x^3 + x^4 + \cdots]^4$

▲ $^9c_8$

⑥. Determine the constant in $\left[3x^2 - \left(\frac{2}{x}\right)\right]^{15}$.

▲ $[X+Y]^{15} = \sum_{r=0}^{15} 15c_r \left[3x^2\right]^r \left(-\frac{2}{x}\right)^{15-r}$

$= \sum_{r=0}^{15} 15c_r \; 3^r x^{2r} (-2)^{15-r} x^{-15+r}$

$= \sum_{r=0}^{15} 15c_r \; 3^r \; x^{3r-15} \; (-2)^{15-r}$

For $r=5$, we have $15c_5 \; 3^5 \; x^0 \; (-2)^{10} = 15c_5 \times 3^5 \times 2^{10}$

7). Find the coefficient of x^7 in $[1+x+x^2+\cdots]^{15}$

$$f(x) = [1+x+x^2+\cdots]^{15} = \left[\frac{1}{1-x}\right]^{15} = [1-x]^{-15}$$

$$= \sum_{r=0}^{\infty} {}^{-15}C_r (-x)^r$$

$$= \sum_{r=0}^{\infty} (-1)^r \, {}^{15+r-1}C_r (-x)^r$$

For $r=7$, we have $(-1)^7 \, {}^{21}C_7 (-x)^7 = {}^{21}C_7 \, x^7$

∴ The coefficient of x^7 is ${}^{21}C_7$.

⑧. Find the coefficient of x^7 in $[1+x+x^2+\cdots]^n$, $n \in z^+$

$$f(x) = \left[\frac{1}{1-x}\right]^n = [1-x]^{-n} = \sum_{r=0}^{\infty} {}^{-n}C_r (-x)^r$$

$$= \sum_{r=0}^{\infty} (-1)^r \, {}^{n+r-1}C_r (-x)^r$$

For $r=7$, we have $(-1)^7 \, {}^{n+6}C_6 [-x]^7 = {}^{n+6}C_6 \, x^7$

∴ The coefficient of x^7 is $\boxed{{}^{n+6}C_6}$

⑨. Find the coefficient of x^{50} in $[x^7+x^8+x^9+\cdots]^6$

$$f(x) = x^{42}[1+x+x^2+\cdots]^6 = x^{42}\left[\frac{1}{1-x}\right]^6 = x^{42} \sum_{r=0}^{\infty} {}^{-6}C_r (-x)^r$$

$$= x^{42} \sum_{r=0}^{\infty} {}^{6+r-1}C_r \, x^r$$

For $r=8$, we have $x^{42} \times {}^{6+8-1}C_8 \times x^8 = {}^{13}C_8 \, x^{50}$

∴ The coefficient of x^{50} is ${}^{13}C_8$.

⑩. Find the coefficient of x^{20} in $[x^2+x^3+x^4+x^5+x^6]^5$.

$$f(x) = x^{10}[1+x+x^2+x^3+x^4]^5$$

$$f(x) = x^{10}\left[\frac{1-x^5}{1-x}\right]^5 = x^{10}[1-x^5]^5 [1-x]^{-5}$$

$$= x^{10} \sum_{r=0}^{5} {}^5C_r [-x^5]^r \sum_{r=0}^{\infty} {}^{-5}C_r [-x]^r$$

$$= x^{10} \sum_{r=0}^{5} (-1)^r \, {}^5C_r \, x^{5r} \sum_{r=0}^{\infty} (-1)^r \, {}^{5+r-1}C_r [x]^r$$

$$= x^{10} \sum_{r=0}^{5} (-1)^r \, {}^5C_r \, x^{5r} \sum_{K=0}^{\infty} {}^{5+K-1}C_K [x]^K$$

When $r=0$; $K=10$

$x^{10} \; {}^{5+10-1}C_{10} \times x^{10} = {}^{14}C_{10} \, x^{20}$

$r=2$; $K=0$

$x^{10} \times {}^5C_2 \, x^{10} = {}^5C_2 \times x^{20}$

$r=1$, $K=5$

$= x^{10}(-1) \, {}^5C_1 \times x^5 \times {}^9C_5 \, x^5$

$= -{}^5C_1 \times {}^9C_5 \times x^{20}$

∴ The coefficient of x^{20} is ${}^{14}C_{10} + {}^5C_2 - {}^5C_1 \times {}^9C_5$

(11) Find the coefficient of x^{24} in the

▶ $f(x) = x^{12}\left[1 + x + \cdots x^5\right]^4 = x^{12}\left[\frac{1-x^6}{1-x}\right]^4 = x^{12}[1-x^6]^4[1-x]^{-4}$

$f(x) = x^{12}\sum_{r=0}^{4} {}^4C_r\left[-x^6\right]^r \sum_{r=0}^{\infty} {}^{-4}C_r(-x)^r$

$f(x) = x^{12}\sum_{r=0}^{4}(-1)^r {}^4C_r\, x^{6r} \sum_{r=0}^{\infty} {}^{4+r-1}C_r\, x^r$

$f(x) = x^{12}\sum_{r=0}^{4}(-1)^r {}^4C_r\, x^{6r} \sum_{k=0}^{\infty} {}^{4+k-1}C_k\, x^k$

when $r=0$; $k=12$

x^{12} ${}^{15}C_{12}\, x^{12} = {}^{15}C_{12}\, x^{24}$

when $r=2$; $k=0$

$x^{12}(-1)^2\, {}^4C_2\, x^{12} = {}^4C_2 \times x^{24}$

when $r=1$; $k=6$

$x^{12}(-1)\, {}^4C_1\, x^6 \times {}^9C_6\, x^6 = (-1)\, {}^4C_1 \times {}^9C_6\, x^{24}$

The coefficient of x^{24} is

${}^{15}C_{12} + {}^4C_2 - \left[{}^4C_1 \times {}^9C_6\right]$ =

(12) Find the coefficient of x^{15} in $x^3[1-2x]^{10}$.

▶ Zero

(13) — 11 — x^{15} in $\frac{x^3 - 5x}{[1-x]^3}$

▶ $f(x) = [x^3 - 5x]\sum_{r=0}^{\infty} {}^{-3}C_r(-x)^r = [x^3 - 5x]\sum_{r=0}^{\infty} {}^{3+r-1}C_r\, x^r$

$f(x) = x^3\sum_{r=0}^{\infty} {}^{3+r-1}C_r\, x^r - 5x\sum_{r=0}^{\infty} {}^{3+r-1}C_r\, x^r$

$f(x) = x^3\sum_{r=0}^{\infty} {}^{3+r-1}C_r\, x^r - 5x\sum_{k=0}^{\infty} {}^{3+k-1}C_k\, x^k$

when $r=12$; $k=14$

$x^3\, {}^{14}C_{12}\, x^{12} - 5x\, {}^{16}C_{14}\, x^{14}$

$= {}^{14}C_{12}\, x^{15} - 5 \times {}^{16}C_{14}\, x^5$ =

∴ The coefficient of x^{15} is $\boxed{{}^{14}C_{12} - 5 \times {}^{16}C_{14}}$

(14) Determine the coefficient of x^8 in $\frac{1}{(x-3)(x-2)^2}$

▶ We have $\frac{1}{(x-3)(x-2)^2} = \frac{A}{x-3} + \frac{B}{x-2} + \frac{C}{(x-2)^2}$

$1 = A(x-2)^2 + B(x-3)(x-2) + C(x-3)$

Put $x=3$, $\boxed{A=1}$

Put $x=2$: $\boxed{C=-1}$

$A+B=0 \Rightarrow B=-A \Rightarrow \boxed{B=-1}$

$$f(x) = \frac{1}{x-3} - \frac{1}{x-2} - \frac{1}{(x-2)^2}$$

$$= -\frac{1}{3}\frac{1}{\left[1-\frac{x}{3}\right]} + \frac{1}{2}\frac{1}{\left[1-\frac{x}{2}\right]} - \frac{1}{4}\frac{1}{\left[1-\frac{x}{2}\right]^2}$$

$$= \left(-\frac{1}{3}\right)\sum_{r=0}^{\infty} {}^{-1}C_r\left(-\frac{x}{3}\right)^r + \frac{1}{2}\sum_{r=0}^{\infty} {}^{-1}C_r\left(-\frac{x}{2}\right)^r - \frac{1}{4}\sum_{r=0}^{\infty} {}^{-2}C_r\left(-\frac{x}{2}\right)^r$$

$$f(x) = \left(-\frac{1}{3}\right)\sum_{r=0}^{\infty}\left(\frac{x}{3}\right)^r + \frac{1}{2}\sum_{r=0}^{\infty}\left(\frac{x}{2}\right)^r - \frac{1}{4}\sum_{r=0}^{\infty} {}^{2+r-1}C_r\left(\frac{x}{2}\right)^r$$

$$f(x) = \left(-\frac{1}{3}\right)\sum_{r=0}^{\infty}\left(\frac{x}{3}\right)^r + \frac{1}{2}\sum_{r=0}^{\infty}\left(\frac{x}{2}\right)^r - \frac{1}{4}\sum_{r=0}^{\infty} {}^{1+r}C_r\left(\frac{x}{2}\right)^r$$

For $r=8$, we have

~~$-\left(\frac{x}{3}\right)^8 + \frac{1}{2}\left(\frac{x}{2}\right)^8 - \frac{1}{4}\,{}^{9}C_8\left(\frac{x}{2}\right)^8$~~

$= \left[\left(\frac{1}{3}\right)\left(\frac{1}{3^8}\right)x^8 + \frac{1}{2}\left(\frac{1}{2^8}\right)x^8 - \frac{9}{4}\left(\frac{1}{2^8}\right)x^8\right]$

$= -\frac{1}{3}\times\left(\frac{1}{3}\right)^8 x^8 + \frac{1}{2}\left(\frac{1}{2}\right)^8 x^8 - \frac{1}{4}\,{}^{9}C_8\left(\frac{1}{2}\right)^8 x^8$

$= \left[-\frac{1}{3}\times\left(\frac{1}{3}\right)^8 + \frac{1}{2}\left(\frac{1}{2}\right)^8 - \frac{9}{4}\left(\frac{1}{2}\right)^8\right]x^8$

$= \left[-\frac{1}{3}\left(\frac{1}{3}\right)^8 - \frac{7}{4}\left(\frac{1}{2}\right)^8\right]x^8$

$= -\left[\left[\frac{1}{3}\right]^9 + 7\left[\frac{1}{2}\right]^{10}\right]x^8$

∴ The coefficient of x^8 is $\boxed{-\left[\left[\frac{1}{3}\right]^9 + 7\left[\frac{1}{2}\right]^{10}\right]}$

(18) Determine the constant [i.e, coefficient of x^0] in $\left[4x^3 - \frac{5}{x}\right]^{16}$.

$$f(x) = \left[4x^3 - \frac{5}{x}\right]^{16} = \sum_{r=0}^{16} {}^{16}C_r \left[4x^3\right]^r \left[-\frac{5}{x}\right]^{16-r}$$

$$= \sum_{r=0}^{16} {}^{16}C_r \, 4^r x^{3r} [-5]^{16-r} x^{r-16}$$

$$= \sum_{r=0}^{16} {}^{16}C_r \, 4^r [-5]^{16-r} x^{4r-16}$$

For $r=4$, we have:

$${}^{16}C_4 \times 4^4 [-5]^{16-4} \times x^0 = {}^{16}C_4 \times 4^4 \times 5^{12} \times x^0$$

$\therefore$ The constant term is $\boxed{{}^{16}C_4 \times 4^4 \times 5^{12}}$

Find the coefficient of x^{60} in $\left[x^8 + x^9 + x^{10} + \cdots\right]^7$.

$$f(x) = x^{56}\left[1 + x + x^2 + \cdots\right]^{-7} = x^{56}\left[\frac{1}{1-x}\right]^{-7} = x^{56}[1-x]^{-7}$$

$$= x^{56} \sum_{r=0}^{\infty} {}^{-7}C_r [-x]^r$$

$$= x^{56} \sum_{r=0}^{\infty} {}^{6+r}C_r \, x^r$$

For $r=4$, we have:

$$x^{56} \, {}^{10}C_4 \times x^4 = {}^{10}C_4 \times x^{60}$$

$\therefore$ The coefficient of x^{60} is ${}^{10}C_4$

(17) Determine the sequence generated by $[1-4x]^{-1/2}$

We have $[1-4x]^{-1/2} = \sum_{r=0}^{\infty} {}^{-1/2}C_r [-4x]^r$

$$= {}^{-1/2}C_0 + {}^{-1/2}C_1 [-4x]^1 + {}^{-1/2}C_2 [-4x]^2 + \cdots$$

The coefficient of x^n is ${}^{-1/2}C_n [-4]^n$

$$= \cdots$$

$$= \cdots$$

$$= \cdots$$

$$= \cdots$$

$$-\tfrac{1}{2}C_n\,[-4]^n = \frac{\left(-\frac{1}{2}\right)\left(-\frac{1}{2}-1\right)\left(-\frac{1}{2}-2\right)\cdots\left(-\frac{1}{2}-n+1\right)}{n!}(-4)^n$$

$$= \frac{\left(1+\frac{2n-2}{2}\right)\cdots\left(\frac{5}{2}\right)\left(\frac{3}{2}\right)\left(\frac{1}{2}\right)}{n!}(4)^n$$

$$= \frac{(2n-1)\cdots 5\cdot 3\cdot 1}{n!}\,2^n\,\frac{n!}{n!}$$

$$= \frac{(2n)!}{n!\,n!}$$

$$= {}^{2n}C_n \text{ , where } n \in N$$

(18) Using the generating function find the number of ways to select 3 books from 6 distinct books.

▶ For each of the books there are two possibilities a book is selected or not, which can be represented as a power series of the form x^0+x^1.

∴ The generating function for the given problem is $= [x^0+x^1]^6 = [1+x]^6 = \sum_{r=0}^{6} {}^6C_r\, x^r$

The coefficient of x^3 is ${}^6C_3$

∴ ${}^6C_3$ ways to select 3 books from 6 distinct books.

(19) In how many ways can 12 marbels be distributed among three students A, B, C, so that A gets at least 4, B and C gets at least 2 so that C gets no more than 5 marbles?

▶ A: $x^4+x^5+x^6+\cdots$

B: $x^2+x^3+x^4+\cdots$

C: $x^2+x^3+x^4+x^5$

$$f(x) = [x^4+x^5+x^6+\cdots]\,[x^2+x^3+x^4+\cdots]\,[x^2+x^3+x^4+x^5]$$

$$= \frac{x^4}{1-x}\times\frac{x^2}{1-x}\times x^2\,[1+x+x^2+x^3]$$

$$= x^8\,\frac{1}{(1-x)^2}\times\frac{1-x^4}{1-x} = x^8[1-x^4]\times\frac{1}{(1-x)^3}$$

$$= x^8[1-x^4]\sum_{r=0}^{\infty} {}^{3+r-1}C_r\, x^r$$

$$= x^8[1-x^4]\sum_{r=0}^{\infty} {}^{2+r}C_r\, x^r$$

$$f(x) = x^8 \sum_{r=0}^{\infty} {}^{2+r}C_r\, x^r - x^{12} \sum_{K=0}^{\infty} {}^{2+K}C_K\, x^K$$

for $r = 4$ and $K = 0$, we have

$${}^{6}C_4\, x^{12} - x^{12} = \left[{}^{6}C_4 - 1\right] x^{12}.$$

$\therefore$ ${}^{6}C_4 - 1$ ways to distribute.

In how many ways we can select 'r' objects with repeatation from 'n' distinct objects?

Power series representation for the selection of each of the n objects is $x^0 + x^1 + x^2 + \cdots$

$\therefore$ The generating function is

$$f(x) = \left[1 + x + x^2 + \cdots\right]^n = \frac{1}{(1-x)^n} = \sum_{r=0}^{\infty} {}^{n+r-1}C_r\, x^r$$

Hence the coefficient of x^r is: $\boxed{{}^{n+r-1}C_r}$

(21) In how many ways can a police captain distribute 24 rifle shells to four police officers so that each officer gets at least three shells, but not more than eight?

The choices for the no. of shells each officer receives are given by $\left[x^3 + x^4 + \cdots + x^8\right]$. There are four officers, so the resulting generating function is

$$f(x) = \left[x^3 + x^4 + \cdots + x^8\right]^4 = x^{12}\left[1 + x + \cdots\right]^4$$

$$f(x) = x^{12}\left[\frac{1-x^6}{1-x}\right]^4 = x^{12}\left[1-x^6\right]^4 \left[1-x\right]^{-4}$$

$$f(x) = x^{12} \sum_{r=0}^{4} {}^{4}C_r\, x^{6r} \underline{(-1)^r} \sum_{K=0}^{\infty} {}^{3+K}C_K\, x^K$$

for $r = 2$: $K = 0$, we have ${}^{4}C_2\, x^{24}$

for $r = 0$: $K = 12$, we have ${}^{15}C_{12}\, x^{24}$

for $r = 1$: $K = 6$, we have $-{}^{4}C_1 \times {}^{9}C_6\, x^{24}$.

The required coefficient is $= {}^{4}C_2 + {}^{15}C_{12} - \left[{}^{4}C_1 \times {}^{9}C_6\right]$

$=$

$S = \{1, 2, \ldots, 15\}$

$S_1 = \{1, 3, 5, 7\} \Rightarrow 0 < \underbrace{1}_{0} < \underbrace{3}_{1} < \underbrace{5}_{1} < \underbrace{7}_{1} < \underbrace{16}_{8} \quad = 11$

$S_2 = \{3, 9, 10, 14\} \quad 0 < 3 < 9 < 10 < 14 < 16$

$2 + 5 + 0 + 3 + 1 = 11$

$S_3 = \{3, 7, 12, 15\} \quad 0 < 3 < 7 < 12 < 15 < 16$

Every subset of S is of the form

$A = \{a_1, a_2, a_3, a_4\}$, where $1 \le a_1 < a_2 < a_3 < a_4 \le 15$,

and $a_2 - a_1 \ge 2$, $a_3 - a_2 \ge 2$ and $a_4 - a_3 \ge 2$.

Let us set $x_1 = a_1 - 1$; $x_2 = a_2 - a_1$; $x_3 = a_3 - a_2$; $x_4 = a_4 - a_3$

$x_5 = 15 - a_4$. Then

$x_1 \ge 0$, $x_2 \ge 2$, $x_3 \ge 2$, $x_4 \ge 2$, $x_5 \ge 0$ — (i)

$x_1 + x_2 + x_3 + x_4 + x_5 = a_1 - 1 + a_2 - a_1 + a_3 - a_2 + a_4 - a_3 + 15 - a_4 = 14$.

$x_1 + x_2 + x_3 + x_4 + x_5 = 14$ — (ii)

Thus, with each subset of A of the required type, $\exists$ a set of integers x_1, x_2, x_3, x_4, x_5 such that the equation (ii) and constraints (i) hold. Hence, the no. of subsets of S of the desired type is equal to the no. of integer solutions of the equation (ii) under the constraint.

$$f(x) = (x^0 + x^1 + x^2 + \cdots) \times [x^2 + x^3 + x^4 + \cdots] \times [x^2 + x^3 + x^4 + \cdots] \times [x^2 + x^3 + \cdots] \times [x^0 + x^1 + x^2 + \cdots]$$

$$= x^6 \frac{1}{(1-x)^2 (1-x)^3}$$

$$= x^6 (1-x)^{-5} = x^6 \sum_{r=0}^{\infty} {}^{4+r}C_r \, x^r$$

The coefficient of x^{14} is $x^6 \times {}^{4+8}C_8 \, x^8$

$= {}^{12}C_8 = 495$ ways.

(23) In how many ways can 3000 identical ... be divided into packs of 25 among 4 groups so that each group gets at least 150 but not more than 1000 envelopes

No. of packs $= \frac{3000}{25} = 120$

Each group should get at least $\frac{150}{25} = 6$ packets

But not more than 1000 envelopes $= \frac{1000}{25} = 40$ packets

$$\therefore f(x) = [x^6 + x^7 + \cdots + x^{40}]^4$$

$$= x^{24}[1 + x^2 + \cdots + x^{34}]^4$$

$$= x^{24}\left[\frac{1-x^{35}}{1-x}\right]^4$$

$$= x^{24}[1-x^{35}]^4[1-x]^{-4}$$

$$= x^{24}[1-x^{35}]^4[1-x]^{-4}$$

$$= x^{24}\sum_{r=0}^{4} {}^4C_r[-x^{35}]^r \sum_{r=0}^{\infty} {}^{3+r}C_r\, x^r$$

$$= x^{24}\sum_{r=0}^{4} (-1)^r {}^4C_r\, x^{35r} \sum_{r=0}^{\infty} {}^{3+r}C_r\, x^r$$

$$f(x) = x^{24}\sum_{r=0}^{4} (-1)^r {}^4C_r\, x^{35r} \sum_{K=0}^{\infty} {}^{3+K}C_K\, x^K$$

For $r=0$; $K=96$

$$x^{120}$$

$${}^{99}C_{96}$$

$r=1$; $K=61$

$$(-1)\,{}^4C_1 \times {}^{64}C_{61}\; x^{120}$$

$r=2$; $K=26$

$${}^4C_2 \times {}^{29}C_{26}\; x^{120}$$

∴ The required coefficient

$$= {}^{99}C_{96} - 4\left({}^{64}C_{61}\right) + {}^4C_2\, {}^{29}C_{26}.$$

A die is rolled for 12 times. what is the probability that the sum of the rolls is 25?

The Power Series for the rolling a die is

$$[x^1+x^2+x^3+x^4+x^5+x^6]$$

Hence the generating function is

$$f(x) = [x+x^2+x^3+x^4+x^5+x^6]^{12}$$

$$= x^{12}\left[\frac{1-x^6}{1-x}\right]^{12}$$

$$\therefore f(x) = x^{12}[1-x^6]^{12}[1-x]^{-12}$$

$$f(x) = x^{12}\sum_{r=0}^{\infty} {}^{12}C_r (-1)^r x^{6r} \sum_{K=0}^{\infty} {}^{11+K}C_K x^K$$

For $r=0$; $K=13$, we have

$$f(x) = {}^{24}C_{13}\, x^{25}$$

$r=1$; $K=7$

$$(-1)\,{}^{12}C_1 [x^{25}]\, {}^{18}C_7 = -{}^{12}C_1 \times {}^{18}C_7\, x^{25}$$

$r=2$; $K=1$

$${}^{12}C_2 \times {}^{12}C_1\, x^{25}$$

$\therefore$ The required coefficient x^{25} is

$$= {}^{24}C_{13} - {}^{12}C_1 \times {}^{18}C_7 + {}^{12}C_2 \times {}^{12}C_1$$

$$\therefore \text{Probability} = \frac{{}^{24}C_{13} - [{}^{12}C_1 \times {}^{18}C_7] + [{}^{12}C_2 \times {}^{12}C_1]}{(6^{12})}$$

(25) Determine the no. of ways to color the squares of a $1 \times n$ chessboard using the colors, red, white and blue, if an even no. of squares are to be colored red.

▲ Let a_n denote the no. of such colorings, with $a_0 = 1$. Let an equal the no. of n-permutations of a multiset of three colors red, white and blue, each with an infinite repetition number in which red occurs an even no. of times. Thus an exponential generating function for the sequence $a_0, a_1, a_2, \ldots$ is the product of red, white and blue factors

$$\therefore f(x) = \left[1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \cdots\right]\left[1 + x + \frac{x^2}{2!} + \cdots\right]\left[1 + x + \frac{x^2}{2!} + \cdots\right]$$

$$= \frac{1}{2}\left(e^x + e^{-x}\right) e^x e^x$$

$$= \frac{1}{2}\left[e^{3x} + e^x\right]$$

$$= \frac{1}{2}\left[\sum_{n=0}^{\infty} 3^n \frac{x^n}{n!} + \sum_{n=0}^{\infty} \frac{x^n}{n!}\right] = \frac{1}{2}\sum_{n=0}^{\infty}\left[3^n + 1\right]\frac{x^n}{n!}$$

$\therefore$ The x^n coeff^n is $\frac{1}{2}\left[3^n + 1\right]$

Hence $a_n = \left[\frac{3^n + 1}{2}\right]$ is the no. of ways to color the chessboard as stated.

(26) In how many ways can two dozen identical robots be assigned to four assembly lines with @ at least 3 robots assigned to each line? (b) at least 3, but not more than nine, robots assigned to each line?

▲ Given that $f(x) = \left[x^3 + x^4 + x^5 + \cdots\right]^4 = x^{12}\left[1 + x + x^2 + \cdots\right]^4$

$$= x^{12}\left[1 - x\right]^{-4} = x^{12}\sum_{r=0}^{\infty} -4C_r\,[-x]^r$$

$$= x^{12}\sum_{r=0}^{\infty} 3+rC_r\, x^r$$

Here, we have to find the coefficient of x^{24}.

For $r = 12$, we have $x^{12} \times {}^{3+12}C_{12}\, x^{12} = {}^{15}C_{12}\, x^{24}$.

$\therefore$ The arrangement of 24 robots of desired type $= {}^{15}C_{12}$

Given that

$$f(x) = \left[x^3 + x^4 + \cdots + x^9\right]^4$$

$$= x^{12}\left[1 + x + x^2 + \cdots + x^6\right]^4$$

$$= x^{12}\left[\frac{1-x^7}{1-x}\right]^4 = x^{12}\left[1-x^7\right]^4\left[1-x\right]^{-4}$$

$$f(x) = x^{12} \sum_{r=0}^{4} {}^4C_r \left[-x^7\right]^r \sum_{k=0}^{\infty} {}^{3+k}C_k \, x^k$$

$$= x^{12} \sum_{r=0}^{4} (-1)^r \, {}^4C_r \, x^{7r} \sum_{k=0}^{\infty} {}^{3+k}C_k \, x^k$$

For $r = 1$; $k = 5$

$$= x^{12}(-1) \times {}^4C_1 \, x^7 \times {}^8C_5 \, x^5 = -{}^4C_1 \times {}^8C_5 \times x^{24}$$

For $r = 0$; $k = 12$

$$x^{12} \, {}^{15}C_{12} \, x^{12} = {}^{15}C_{12} \, x^{24}$$

$\therefore$ The coefficient of x^{24} is $= \left[-{}^4C_1 \times {}^8C_5 + {}^{15}C_{12}\right]$

(27). Two cases of soft drinks, 24 bottles of one type and 24 of another are distributed among five surveyors who are conducting taste tests. In how many ways can the 48 bottles be distributed so that each surveyor gets (a) at least two bottles of each type?
(b) at least two bottles of one particular type and at least three of the other?

(i) The generating function for 24 bottles of type one is

$$f(x) = \left[x^2 + x^3 + \cdots\right]^5 = x^{10}\left[1 + x + x^2 + \cdots\right]^5 = x^{10}\left[1-x\right]^{-5}$$

$$f(x) = x^{10} \sum_{r=0}^{\infty} {}^{4+r}C_r \, x^r$$

For $r = 14$, we have $x^{10} \, {}^{18}C_{14} \, x^{14} = {}^{18}C_{14} \, x^{24}$ one type of

$\therefore$ ${}^{18}C_{14}$ ways can distribute the 24 bottles of soft drink among 5 surveyors so that each gets at least two bottles. Since there are two types, the two cases can be distributed the given restrictions in $\left[{}^{18}C_{14}\right]^2$ ways.

(ii) The generating function for 24 bottles among 5 surveyors so each gets at least three is

$$f(x) = [x^3 + x^4 + x^5 + \cdots]^5 = x^{15}[1 + x + x^2 + \cdots]^5$$

$$f(x) = x^{15}[1-x]^{-5} = x^{15} \sum_{r=0}^{\infty} {}^{4+r}C_r \, x^r$$

For $r = 9$, we have:

$$x^{15} \times {}^{13}C_9 \, x^9 = {}^{13}C_9 \, x^{24}.$$

$\therefore$ ${}^{18}C_{14} \times {}^{13}C_9$ ways can be distributed 48 bottles so that at least two bottles of one type and at least three of the other among 5 surveyors.

Convolution of Sequences :

Consider two sequences $\langle a_r \rangle$ and $\langle b_r \rangle$ generated by the functions $f(x)$ and $g(x)$ respectively, so that

$$f(x) = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + \cdots$$

and $$g(x) = b_0 + b_1 x + b_2 x^2 + b_3 x^3 + \cdots$$

The function $h(x) = f(x)\, g(x)$

$$= [a_0 + a_1 x + a_2 x^2 + a_3 x^3 + \cdots] \times [b_0 + b_1 x + b_2 x^2 + b_3 x^3 + \cdots]$$

$$= a_0 b_0 + [a_0 b_1 + a_1 b_0] x + [a_0 b_2 + a_1 b_1 + a_2 b_0] x^2 + \cdots$$

$\therefore$ $h(x)$ is the generating function for the sequence

$c_0, c_1, c_2, \cdots$

we have $c_k = a_0 b_k + a_1 b_{k-1} + a_2 b_{k-2} + \cdots + a_{k-2} b_2 + a_{k-1} b_1 + a_k$

$$= \sum_{i=0}^{k} a_i \, b_{k-i}$$

Here, the sequence $\langle c_r \rangle$ is called the convolution of the sequence $\langle a_r \rangle$ and $\langle b_r \rangle$.